

Erdős Problem #51

EP-51

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Is there an infinite set $A \subset \mathbb{N}$ such that for every $a \in A$ there is an integer n such that $\phi(n) = a$, and yet if n_a is the smallest such integer then $n_a/a \rightarrow \infty$ as $a \rightarrow \infty$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>